

IDS Pulse Operations Suite — QA Testing Guide for Diana

This guide outlines how to perform a complete, end-to-end audit and functional test of the **IDS Pulse Operations Suite**. The system uses a hybrid sync architecture: actions taken on the mobile app simulator are immediately written to local cache and synchronized with the live PostgreSQL database on Supabase.

Step 1: Authentication & Access

1. Open the live system: `[proud-lavoisier.vercel.app](https://proud-lavoisier.vercel.app)`
2. In the central login panel, authenticate with your designated admin credentials:
 - **Username / Email:** `diana`
 - **Password:** `DianaPulse2026!
3. Click **Unlock Workspace**. The system will authenticate your session and land you directly on the **Projects Registry** tab.

Note: Your administrator account has `owner` privileges, granting you read and write access across all tabs.

Step 2: Onboard Real QA Test Data

To test system workflows under real operational scenarios, register a new client company, map a manufacturing plant, and set up a representative:

A. Register a Client (Supplier)

1. Go to the **Setup & Onboarding** tab in the sidebar.
2. In the **Supplier Directory** sub-tab, scroll to **Register New Client/Supplier**.
3. Enter details:
 - **Supplier Name:** e.g., `BorgWarner`
 - **Contact Name & Email:** e.g., `Sarah Connor` (`s.connor@borgwarner.com`)
4. Click **Create Supplier**. The client will persist in Supabase and display in the registry.

B. Map a Plant & Location

1. Switch to the **Plants & Locations** sub-tab.
2. Under **Map New Plant/Location**:
 - **Location Name:** e.g., `BorgWarner Cadillac`
 - **Address:** e.g., `123 Plant Road, Cadillac, MI`
 - **Primary Representative:** Select a representative (e.g., `Hugo Picon`)
 - **Standard Weekly Hours:** e.g., `45`
 - **Custom Bill Rate:** e.g., `42.50`
 - **Mapped Supplier:** Select `BorgWarner` (created in Step 2A)
3. Click **Map Location**. This maps the location and creates default billing rates.

Step 3: Dispatch Daily Tasks

Verify the real-time representative dispatch planner:

1. Go to the **Daily Tasks Planner** tab in the sidebar.
2. Under **Dispatch Representative Checklist**:
 - **Representative:** Select `Hugo Picon`
 - **Task Instruction:** e.g., `Inspect containment racks for fuel pump harness connectors. Report any loose pins.`
3. Click **Dispatch Task**. The task is written to Supabase and immediately pushed to the representative's screen.

Step 4: Simulate Field Rep Activities (Mobile Panel)

Verify zero-latency data entry and drawing annotations using the simulated mobile screen:

1. Look at the left panel labeled **"Clarence's Phone"**.
2. Log out if logged in, and enter a passcode to log in as a field representative:
 - **Passcode:** `hugo` (to log in as Hugo Picon, whom we dispatched in Step 3).
3. Review the Handover Lock screen showing recent plant events, and click **Acknowledge Handover**.
4. Click **Start Work Shift**. You will immediately see the custom checklist task dispatched in Step 3.
5. Tap **New Suspect Material** to test the 4-step Incident Report wizard:
 - **Step 1 (Photos):** Capture/upload a photo. Use the canvas drawing tools to draw a **red circle** highlighting the defect area on the photo, then click Next.
 - **Step 2 (Traceability Scan):** Click **Scan Barcode** to simulate scanning a factory container. It will parse Part Numbers and Bins automatically. Enter quantity `5` and click Add Part.
 - **Step 3 (Describe):** Select classification `PRR`, enter a description (e.g., `Bent connector pins`), and list actions taken.
 - **Step 4 (Review):** Review the summary and click **Submit Incident**.
6. Tap **Log Rework** to record sort operations:
 - Select Part Number, enter quantity `12`, time spent `15 minutes`, and click Save.
7. Click **End Work Shift** to complete the representative shift checklist and submit the Shift Walkthrough.

Step 5: Real-Time Synchronization & Audit

1. Go back to the manager **CRM Dashboard** on the right side.
2. In the top header bar, you will receive real-time audio alerts and toast notifications as the mobile simulator uploads logs.
3. Select the **Incidents & Shift Logs** tab.
4. Under **Suspect Material Logs**, click on the newly created BorgWarner incident. Verify that the details, parts list, and your annotated defect photo are correctly loaded.
5. Select the **Rework Feed** tab to verify the rework quantities are logged.
6. Verify the **Timesheets & Mileage** tab to see that Hugo's shift hours and mileage have been automatically recorded.

Step 6: Financial Operations & Report Exports

Validate reporting and invoicing calculations:

1. Select the **Billing & Invoices** tab.
2. Under **Generate Supplier Invoice**:
 - **Supplier:** Select `BorgWarner`
 - **Currency:** Select `USD`
3. Review the calculated totals (Hours worked x Billing Rate).
4. Click **QuickBooks CSV** to export QuickBooks-formatted timesheets.
5. Click **Download PDF Report** to generate a clean, customer-ready PDF statement. Verify that long descriptions wrap correctly and do not overflow into other columns.